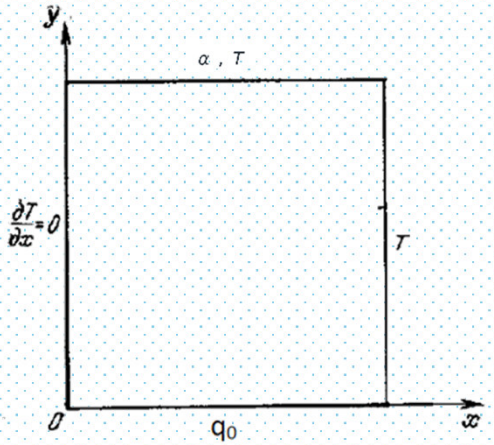
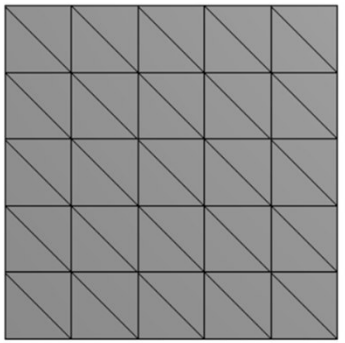
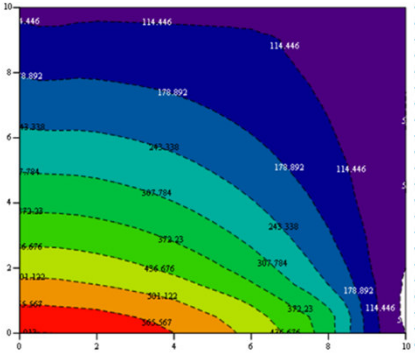
Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

Finite Element Analysis of the Steady-State Thermal Field of a Plate in a Plane Formulation



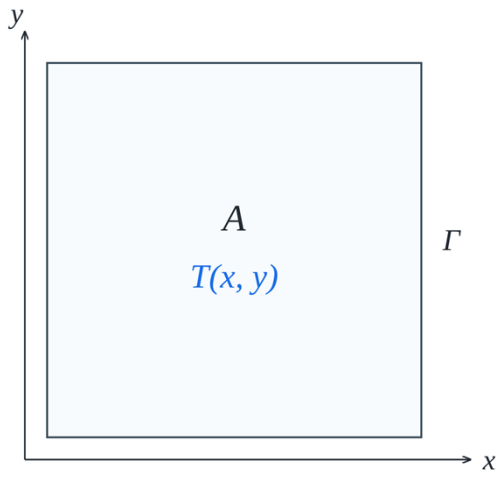


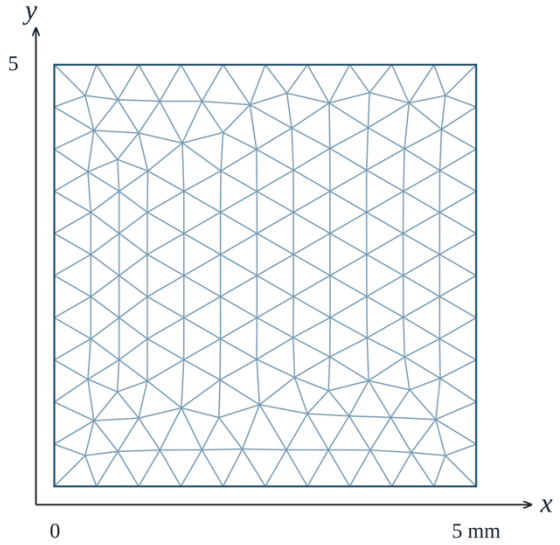


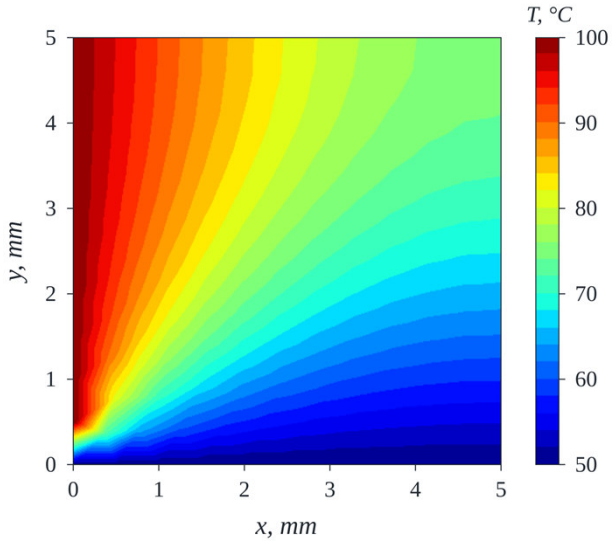
Teachers: Malastowski Nikolay, Blinov Artem

Problem statement

Steady-state temperature field in a flat plate







Computational domain **Finite element mesh** **Temperature field**

$$T = T(x, y), \qquad h = \text{const}, \qquad \lambda = \text{const} > 0$$

T: temperature field; h: plate thickness; λ: thermal conductivity.

Goal: find the nodal temperatures and recover the field inside the plate.

Three-dimensional heat conduction equation

1. Heat balance

$$\int_V \rho c \frac{\partial T}{\partial t}\, dV = -\int_{\partial V} \mathbf{q}\cdot\mathbf{n}\, dS + \int_V Q\, dV$$

Heat accumulation = boundary heat input
+ internal heat generation.

2. Local governing equation

$$\rho c \frac{\partial T}{\partial t} = -\nabla\cdot\mathbf{q} + Q$$

Position vector: $\mathrm{x} = (\mathrm{x}, \mathrm{y}, \mathrm{z})^{\mathrm{T}}$.
ρ: density; c: specific heat capacity.
Q: volumetric heat generation rate.

3. Fourier's law

$$\mathbf{q} = -\mathbf{\Lambda}\nabla T$$

$$\rho c \frac{\partial T}{\partial t} = \nabla\cdot(\mathbf{\Lambda}\nabla T) + Q$$

4. Isotropic material

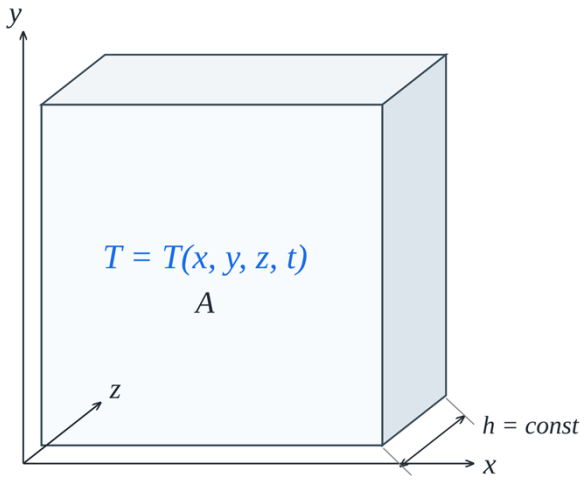
$$\mathbf{\Lambda} = \lambda\mathbf{I} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$\rho c \frac{\partial T}{\partial t} = \nabla\cdot(\lambda\nabla T) + Q$$

Λ is a tensor; λ is the same in every direction.

Isotropy does not imply λ = const. Specify the initial field and boundary conditions separately.

Plate geometry and Cartesian coordinates



1. Coordinates and thickness

x, y: in-plane coordinates;
z: through-thickness coordinate;
h: constant thickness.

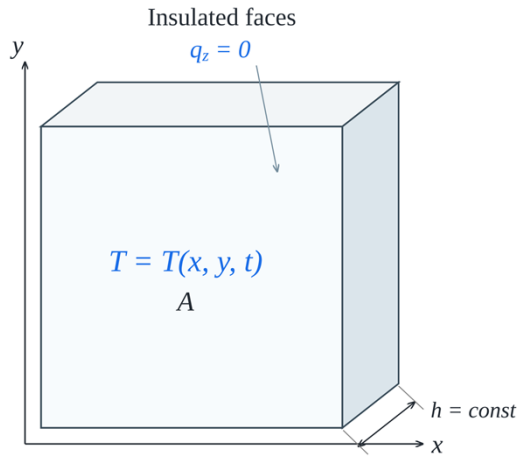
2. General heat balance

The material is isotropic. We still retain three coordinates, time, and source Q.

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(\lambda \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda \frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(\lambda \frac{\partial T}{\partial z}\right) + Q$$

The two-dimensional model follows from assuming that temperature is independent of z.

Temperature is uniform through the thickness



1. Conditions for a two-dimensional model

The geometry, properties, and thermal loads
do not cause temperature to vary with z.
The two broad faces are thermally insulated.

2. No through-thickness derivative

$$T = T(x, y, t), \qquad \frac{\partial T}{\partial z} = 0, \qquad q_z = 0$$

3. Two-dimensional heat balance

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(\lambda \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda \frac{\partial T}{\partial y}\right) + Q$$

A two-dimensional model need not be steady-state: time dependence is still retained.

Two-dimensional steady-state heat conduction

Steady-state model with constant thermal conductivity

1. Assumptions

Steady-state conditions and a homogeneous, isotropic material.

$$\frac{\partial T}{\partial t} = 0, \qquad \lambda = \text{const} > 0$$

2. Heat balance

$$\frac{\partial}{\partial x}\left(\lambda \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda \frac{\partial T}{\partial y}\right) + Q = 0$$

Temperature T = T(x, y). Source Q is retained.

3. Constant λ outside the derivatives

$$\lambda\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + Q = 0$$

There is no heat accumulation over time.

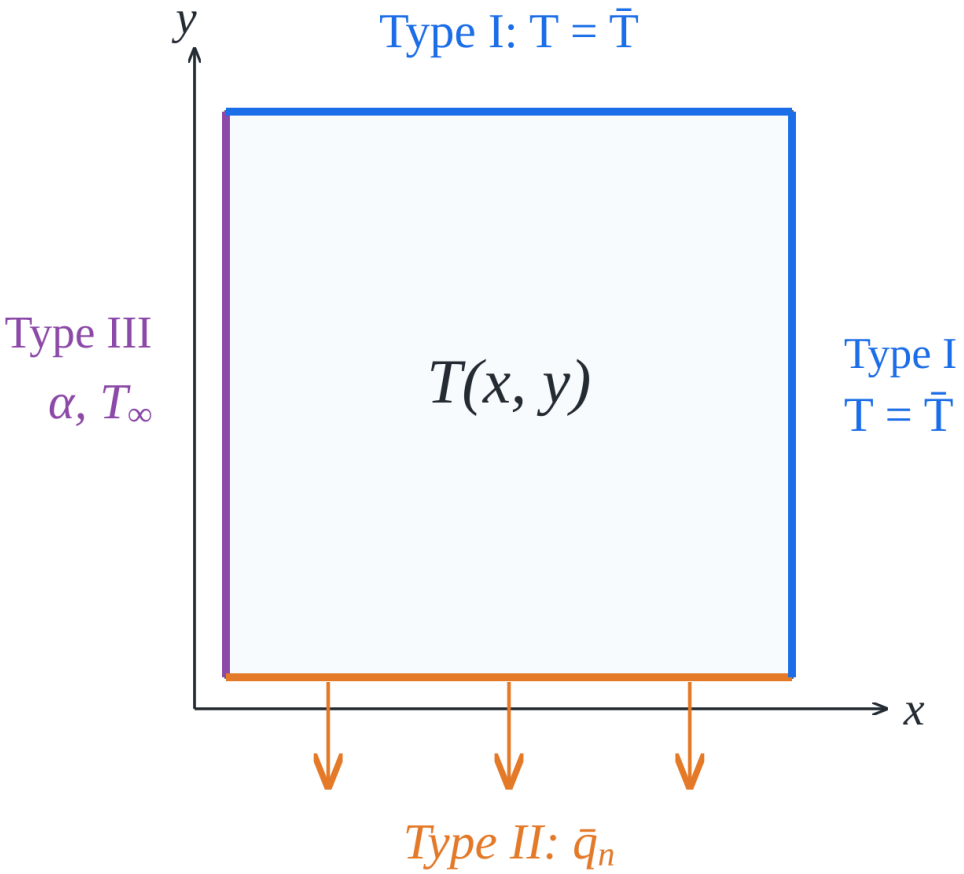
4. Constant thickness

$$dV = h\,dA, \qquad dS_{\text{lat}} = h\,dL$$

A factor h appears in the volume integrals and lateral boundary integrals.

Thickness h accounts for the material volume and the lateral heat-transfer surface area.

Thermal boundary conditions



The normal n points out of the plate

$$q_n = \mathbf{q} \cdot \mathbf{n} = -\lambda \frac{\partial T}{\partial n}$$

Type I. Prescribed temperature

$$T = \overline{T}$$

The boundary temperature is known.

Type II. Prescribed normal heat flux

$$-\lambda \frac{\partial T}{\partial n} = \bar{q}_n$$

$\bar{q}_n > 0$: heat removal; $\bar{q}_n = 0$: insulation.

Type III. Convection to the surroundings

$$-\lambda \frac{\partial T}{\partial n} = \alpha (T - T_\infty)$$

For Type III, α and T∞ are known. The solution determines the surface temperature T.

Variational formulation

1. A number associated with each field

A functional assigns a number
to a temperature field.

$$T(x,y) \longrightarrow \Phi[T]$$

2. The field that minimizes the functional

$$\Phi[T^*] = \min_{T \text{ admissible}} \Phi[T]$$

Admissible fields already satisfy Type I conditions.
We minimize the functional, not temperature.

3. The original thermal problem is retained

Minimizing this functional yields the original
heat conduction equation and the
Type II and Type III boundary conditions.

$$\delta\Phi = 0 \iff \text{original problem}$$

4. FEM: nodal temperatures as unknowns

$$\min \Phi_h \implies \mathbf{KT} = \mathbf{F}$$

After imposing the prescribed temperatures,
we obtain a system of linear algebraic equations.
We solve it and recover the temperature field.

Finite element discretization yields the system of algebraic equations.

Heat conduction functional

General functional: four physical contributions

1. Functional over the volume V

Linear steady-state problem: $\lambda > 0$, $\alpha \geq 0$; input data do not depend on the unknown temperature.

$$\Phi[T] = \int_V \left[\frac{\lambda}{2}|\nabla T|^2 - QT\right] dV + \int_{S_2} \bar{q}_n T \, dS + \frac{1}{2}\int_{S_3} \alpha (T - T_\infty)^2 \, dS$$

2. Meaning of the terms

λ: thermal conductivity within the material;
Q: volumetric heat generation rate.

$\bar{q}_n$: prescribed outward heat flux;
α, $T\infty$: convection parameters.

The Type I condition $T = \bar{T}$ constrains the admissible temperature field.

The plus sign before the prescribed flux corresponds to heat flow out of the body.

Functional for a flat plate

1. Integration over the constant thickness

A: planar domain; Γ_2 and Γ_3: boundary segments with Type II and Type III conditions.

$$dA = dx\,dy, \qquad dV = h\,dA, \qquad dS_{\text{lat}} = h\,dL$$

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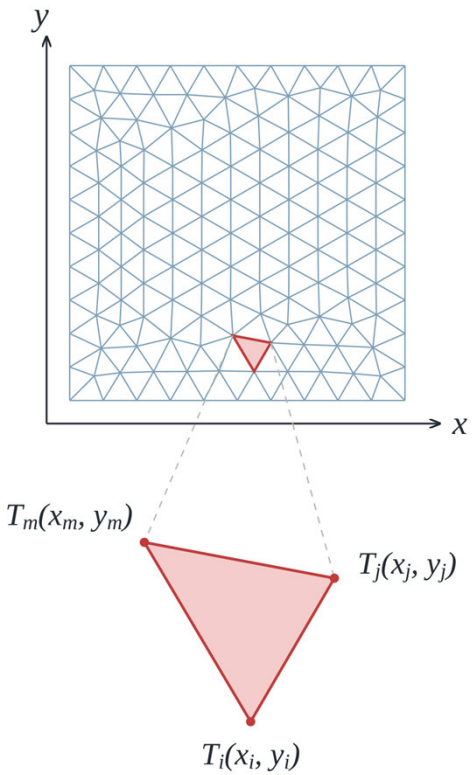
2. All terms in Cartesian coordinates x, y

$$\Phi[T] = h\int_A \left[\frac{\lambda}{2}\left(\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2\right) - QT\right] dA$$
$$+ h\int_{\Gamma_2} \overline{q}_n T\,dL + \frac{h}{2}\int_{\Gamma_3} \alpha(T - T_\infty)^2\,dL.$$

Type I conditions define admissible fields. Heat generation, heat flux, and convection remain.

Temperature approximation from nodal values

Linear approximation within a triangle



1. Polynomial approximation of temperature within a finite element

$$T^{(e)}(x, y) = \beta_0 + \beta_x x + \beta_y y$$

2. Three coefficients to determine

C_e: coordinate matrix of the three vertices:

$$\begin{pmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_m & y_m \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_x \\ \beta_y \end{pmatrix} = \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

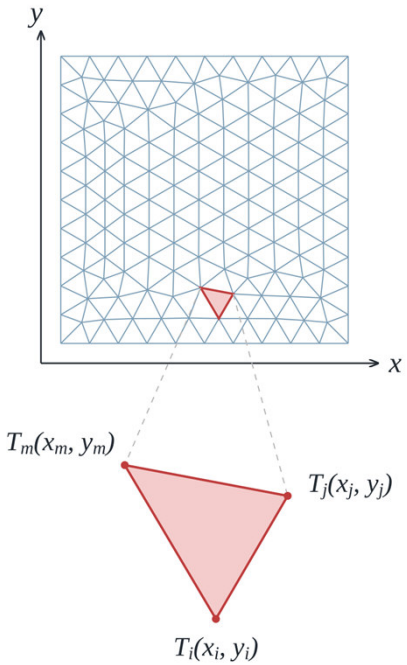
3. Nondegeneracy check

$$D_e = \det \mathbf{C}_e \neq 0, \qquad A_e = \frac{|D_e|}{2}$$

Three nodal temperatures fully determine the linear field within the triangle.

Temperature approximation from nodal values

Polynomial coefficients and shape functions



1. Coefficients in terms of nodal temperatures

$$\beta_0 = \frac{a_i T_i + a_j T_j + a_m T_m}{D_e},$$

$$\beta_x = \frac{b_i T_i + b_j T_j + b_m T_m}{D_e}, \qquad \beta_y = \frac{c_i T_i + c_j T_j + c_m T_m}{D_e}.$$

2. Coefficients depend on vertex coordinates

$$a_i = x_j y_m - x_m y_j, \qquad b_i = y_j - y_m, \qquad c_i = x_m - x_j,$$

$$a_j = x_m y_i - x_i y_m, \qquad b_j = y_m - y_i, \qquad c_j = x_i - x_m,$$

$$a_m = x_i y_j - x_j y_i, \qquad b_m = y_i - y_j, \qquad c_m = x_j - x_i.$$

3. Collecting terms in T_i, T_j, T_m

$$T^{(e)} = \frac{a_i + b_i x + c_i y}{D_e} T_i + \frac{a_j + b_j x + c_j y}{D_e} T_j + \frac{a_m + b_m x + c_m y}{D_e} T_m$$

$$T^{(e)} = N_i T_i + N_j T_j + N_m T_m, \qquad N_a = \frac{a_a + b_a x + c_a y}{D_e}, \quad a \in \{i, j, m\}$$

Shape functions relate the temperature inside an element to its nodal temperatures.
At its own node, N = 1; at the other two nodes, N = 0.

Element functional in terms of nodal temperatures

1. Substitution of temperature interpolation

$$T^{(e)} = N_i T_i + N_j T_j + N_m T_m$$

2. Complete element functional

$$\Phi_e = \frac{\lambda h}{2} \int_{A_e} \left[\left(\sum_a \frac{\partial N_a}{\partial x} T_a \right)^2 + \left(\sum_a \frac{\partial N_a}{\partial y} T_a \right)^2 \right] dA - h \int_{A_e} Q(N_i T_i + N_j T_j + N_m T_m)\, dA$$

$$+ h \int_{\Gamma_{2,e}} \bar{q}_n (N_i T_i + N_j T_j + N_m T_m)\, dL + \frac{h}{2} \int_{\Gamma_{3,e}} \alpha (N_i T_i + N_j T_j + N_m T_m - T_\infty)^2\, dL.$$

Four physical contributions:

$$\Phi_e = \Phi_{\lambda,e} + \Phi_{Q,e} + \Phi_{q,e} + \Phi_{\alpha,e}$$

Expanded sums of derivatives with respect to x and y:

$$\sum_a \frac{\partial N_a}{\partial x} T_a = \frac{\partial N_i}{\partial x} T_i + \frac{\partial N_j}{\partial x} T_j + \frac{\partial N_m}{\partial x} T_m \qquad \sum_a \frac{\partial N_a}{\partial y} T_a = \frac{\partial N_i}{\partial y} T_i + \frac{\partial N_j}{\partial y} T_j + \frac{\partial N_m}{\partial y} T_m$$

Internal mesh edges do not enter the boundary integrals.
The prescribed temperature is satisfied on Γ_1.

Shape-function derivatives and matrix B_e

1. Spatial derivatives of the shape functions

$$N_a = \frac{a_a + b_a x + c_a y}{D_e}, \qquad \frac{\partial N_a}{\partial x} = \frac{b_a}{D_e}, \qquad \frac{\partial N_a}{\partial y} = \frac{c_a}{D_e}, \qquad a = i, j, m$$

2. Derivatives of the temperature interpolation

$$\frac{\partial T^{(e)}}{\partial x} = \frac{\partial N_i}{\partial x} T_i + \frac{\partial N_j}{\partial x} T_j + \frac{\partial N_m}{\partial x} T_m = \frac{b_i T_i + b_j T_j + b_m T_m}{D_e},$$

$$\frac{\partial T^{(e)}}{\partial y} = \frac{\partial N_i}{\partial y} T_i + \frac{\partial N_j}{\partial y} T_j + \frac{\partial N_m}{\partial y} T_m = \frac{c_i T_i + c_j T_j + c_m T_m}{D_e}.$$

3. Matrix form of the derivatives

$$\begin{pmatrix} \partial T^{(e)}/\partial x \\ \partial T^{(e)}/\partial y \end{pmatrix} = \underbrace{\frac{1}{D_e} \begin{pmatrix} b_i & b_j & b_m \\ c_i & c_j & c_m \end{pmatrix}}_{\mathbf{B}_e} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}, \qquad \nabla T^{(e)} = \mathbf{B}_e \mathbf{T}_e$$

The 2×3 matrix B_e relates three nodal temperatures to two derivatives.
The temperature gradient is constant within the element.

Derivation of the functional: heat conduction

1. Original contribution

Thermal conductivity is constant: $\lambda = \text{const} > 0$.

$$\Phi_{\lambda,e} = \frac{\lambda h}{2} \int_{A_e} \left[\left(\frac{\partial T^{(e)}}{\partial x} \right)^2 + \left(\frac{\partial T^{(e)}}{\partial y} \right)^2 \right] dA$$

2. Substitution of the derivatives

$$\Phi_{\lambda,e} = \frac{\lambda h}{2D_e^2} \int_{A_e} \left[(b_i T_i + b_j T_j + b_m T_m)^2 + (c_i T_i + c_j T_j + c_m T_m)^2 \right] dA.$$

3. Evaluation of the area integral

$$\int_{A_e} 1 \, dA = A_e, \qquad D_e^2 = 4A_e^2$$

$$\frac{\lambda h A_e}{2D_e^2} = \frac{\lambda h}{8A_e}$$

4. Sum of squares

$$\Phi_{\lambda,e} = \frac{\lambda h}{8A_e} \left[(b_i T_i + b_j T_j + b_m T_m)^2 + (c_i T_i + c_j T_j + c_m T_m)^2 \right].$$

5. Expansion of the squares

$$\Phi_{\lambda,e} = \frac{\lambda h}{8A_e} \left[(b_i^2 + c_i^2) T_i^2 + (b_j^2 + c_j^2) T_j^2 + (b_m^2 + c_m^2) T_m^2 + 2(b_i b_j + c_i c_j) T_i T_j + 2(b_i b_m + c_i c_m) T_i T_m + 2(b_j b_m + c_j c_m) T_j T_m \right].$$

Spatial integration gives a scalar expression in T_i, T_j, T_m.
Minimization with respect to nodal temperatures follows next.

Functional derivation: volumetric heat source

1. Original term

The source is constant within the element: Q_e = const.

$$\Phi_{Q,e} = -Q_e h \int_{A_e} T^{(e)}\, dA$$

2. Temperature substitution

$$\Phi_{Q,e} = -Q_e h \int_{A_e} (N_i T_i + N_j T_j + N_m T_m)\, dA$$

Take nodal temperatures outside the integral.

3. Triangle centroid

$$x_c = \frac{x_i + x_j + x_m}{3}, \qquad y_c = \frac{y_i + y_j + y_m}{3}$$

For a linear function, the integral equals:

$$\int_{A_e} N_j(x, y)\, dA = A_e N_j(x_c, y_c)$$

4. Shape function value at the centroid

$$N_j(x_c, y_c) = (a_j + b_j x_c + c_j y_c)/D_e$$
$$= \left[N_j(x_i, y_i) + N_j(x_j, y_j) + N_j(x_m, y_m)\right]/3$$
$$= (0 + 1 + 0)/3 = 1/3$$

5. All three integrals are equal

$$\int_{A_e} N_i\, dA = \int_{A_e} N_j\, dA = \int_{A_e} N_m\, dA = \frac{A_e}{3}$$

$$\Phi_{Q,e} = -\frac{Q_e h A_e}{3}(T_i + T_j + T_m)$$

The source gives a term linear in the nodal temperatures.
The factor $Q_e h A_e$ is the total heat generation power in the element.

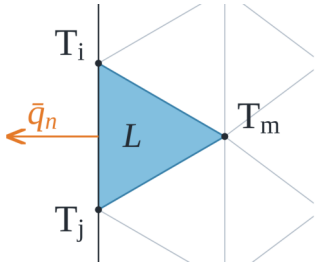
Functional derivation: prescribed heat flux

1. Original boundary term

The flux $\bar{q}_n$ = const is positive outward.

$$\Phi_{q,\gamma} = h\bar{q}_n \int_\gamma T^{(e)}\, dL$$

2. Temperature on the straight edge i, j



$$T^{(e)}(x,y) = N_i(x,y)T_i + N_j(x,y)T_j$$

$$N_i = \frac{a_i + b_i x + c_i y}{D_e}$$

$$N_j = \frac{a_j + b_j x + c_j y}{D_e}, \qquad N_m = 0$$

L: edge length; dL: differential edge length.

3. Boundary edge midpoint

$$x_c = \frac{x_i + x_j}{2}, \qquad y_c = \frac{y_i + y_j}{2}$$

$$N_i(x_c, y_c) = \frac{1+0}{2} = \frac{1}{2}$$

$$N_j(x_c, y_c) = \frac{0+1}{2} = \frac{1}{2}$$

4. Linear shape function integrals along the edge

$$\int_\gamma N_i(x,y)\, dL = L\, N_i(x_c, y_c) = \frac{L}{2}$$

$$\int_\gamma N_j(x,y)\, dL = L\, N_j(x_c, y_c) = \frac{L}{2}$$

5. Substitution of the evaluated integrals

$$\Phi_{q,\gamma} = h\bar{q}_n \left(T_i \frac{L}{2} + T_j \frac{L}{2} \right) = \frac{\bar{q}_n h L}{2}(T_i + T_j)$$

A linear function integrates exactly to L times its value at the midpoint of a straight edge.
The plus sign denotes outward flux. For inward flux $q_{in} > 0$, use $\bar{q}_n = -q_{in}$.

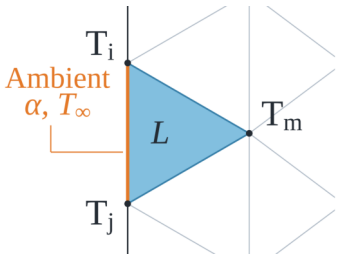
Functional derivation: convection

1. Original Robin boundary term

α and T∞ are constant along the edge.

$$\Phi_{\alpha,\gamma} = \frac{\alpha h}{2} \int_{\gamma} \left(T^{(e)} - T_{\infty} \right)^2 dL$$

2. Interpolation on edge i, j



$$T^{(e)}(x,y) = N_i T_i + N_j T_j$$

$$N_i(x,y) = \frac{a_i + b_i x + c_i y}{D_e}$$

$$N_j(x,y) = \frac{a_j + b_j x + c_j y}{D_e}$$

$$N_m = 0 \quad \text{on edge } i,j$$

3. Temperature substitution

$$\Phi_{\alpha,\gamma} = \frac{\alpha h}{2} \int_{\gamma} \left[N_i T_i + N_j T_j - T_{\infty} \right]^2 dL$$

Here $N_i = N_i(x, y)$, $N_j = N_j(x, y)$.

4. Expansion of the square

$$(N_i T_i + N_j T_j - T_{\infty})^2 = N_i^2 T_i^2 + 2 N_i N_j T_i T_j + N_j^2 T_j^2 - 2 T_{\infty} N_i T_i - 2 T_{\infty} N_j T_j + T_{\infty}^2.$$

5. Expanded expression under the integral

$$\Phi_{\alpha,\gamma} = \frac{\alpha h}{2} \int_{\gamma} \left[N_i^2 T_i^2 + 2 N_i N_j T_i T_j + N_j^2 T_j^2 - 2 T_{\infty} (N_i T_i + N_j T_j) + T_{\infty}^2 \right] dL$$

Evaluate all integrals first. Retain the constant term containing T_{∞}^2 for now.

Convection: evaluation of edge integrals

1. Exact integration using endpoints and midpoint

$$x_c = \frac{x_i + x_j}{2}, \qquad y_c = \frac{y_i + y_j}{2}$$

For f(x, y) quadratic along a straight edge:

$$\int_\gamma f(x,y)\, dL = \frac{L}{6}[f(x_i,y_i) + 4f(x_c,y_c) + f(x_j,y_j)]$$

2. Squared shape functions

$$\int_\gamma N_i^2\, dL = \frac{L}{6}\left[1 + 4(1/2)^2 + 0\right] = \frac{L}{3}$$

$$\int_\gamma N_j^2\, dL = \frac{L}{6}\left[0 + 4(1/2)^2 + 1\right] = \frac{L}{3}$$

3. Product and linear-function integrals

$$\int_\gamma N_i N_j\, dL = \frac{L}{6}[0 + 4(1/2)(1/2) + 0] = \frac{L}{6}$$

$$\int_\gamma N_i\, dL = \int_\gamma N_j\, dL = \frac{L}{2}, \qquad \int_\gamma 1\, dL = L$$

4. Substitution of the evaluated integrals

$$\Phi_{\alpha,\gamma} = \frac{\alpha h}{2}\left[\frac{L}{3}T_i^2 + 2\frac{L}{6}T_iT_j + \frac{L}{3}T_j^2 - 2T_\infty\frac{L}{2}(T_i + T_j) + LT_\infty^2\right]$$

5. Explicit scalar expression

$$\Phi_{\alpha,\gamma} = \frac{\alpha h L}{6}\left[T_i^2 + T_iT_j + T_j^2 - 3T_\infty(T_i + T_j) + 3T_\infty^2\right]$$

For squares and products, use endpoint and midpoint values.
The integrals are exact for a straight edge with constant α and T∞.

Complete scalar functional of a plate element

1. Sum of the evaluated physical contributions

$$\Phi_e(\mathbf{T}_e) = \frac{\lambda h}{8A_e}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right] - \frac{Q_e h A_e}{3}(T_i + T_j + T_m)$$

$$+ \sum_{\gamma=(p,s)\in E_{2,e}} \frac{\bar{q}_{n,\gamma} h L_\gamma}{2}(T_p + T_s) + \sum_{\gamma=(p,s)\in E_{3,e}} \frac{\alpha_\gamma h L_\gamma}{6}\left[T_p^2 + T_pT_s + T_s^2 - 3T_{\infty,\gamma}(T_p + T_s) + 3T_{\infty,\gamma}^2\right].$$

$E_{2,e}$ and $E_{3,e}$: sets of element boundary edges with Type II and Type III conditions.

2. Sum of all element functionals

$$\Phi_{\mathrm{FE}}(\mathbf{T}) = \sum_{e=1}^{n_e} \Phi_e(\mathbf{T}_e)$$

At nodes on Γ_1, retain the constraint $T_a = \bar{T}_a$.

This is a scalar function of nodal temperatures, not a system of equations.
All spatial integrals have already been evaluated.

Functional minimization

1. Minimum of the assembled functional

Keep prescribed temperatures fixed. Minimize over the unknowns:

$$\Phi(\mathbf{T}) = \sum_e \Phi_e(\mathbf{T}_e), \qquad \frac{\partial \Phi}{\partial T_A} = 0$$

2. Nodal equation sums adjacent element contributions

$$\frac{\partial \Phi}{\partial T_A} = \sum_{e:\, A \in e} \frac{\partial \Phi_e}{\partial T_{a(A)}} = 0$$

a(A): local index of global node A. A shared node has one temperature.

3. The derivative of a sum is the sum of derivatives

$$\frac{\partial \Phi_e}{\partial T_a} = \frac{\partial \Phi_e^{(\lambda)}}{\partial T_a} + \frac{\partial \Phi_e^{(Q)}}{\partial T_a} + \frac{\partial \Phi_e^{(q)}}{\partial T_a} + \frac{\partial \Phi_e^{(\alpha)}}{\partial T_a}$$

4. The calculation order is unchanged

$$\lambda \longrightarrow Q \longrightarrow q \longrightarrow \alpha$$

The integrals are already evaluated. Nodal polynomial derivatives give matrices and load vectors.

The global sum is zero, not the derivatives of individual elements or terms.

Minimization: derivative of the conduction contribution

1. Scalar expression

$$\Phi_{\lambda,e} = \frac{\lambda h}{8A_e}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right]$$

2. Derivative with respect to T_a, a = i, j, m

$$\frac{\partial \Phi_{\lambda,e}}{\partial T_a} = \frac{\lambda h}{4A_e}\left[b_a(b_iT_i + b_jT_j + b_mT_m) + c_a(c_iT_i + c_jT_j + c_mT_m)\right]$$

3. Expanded derivative with respect to T_i

$$\frac{\partial \Phi_{\lambda,e}}{\partial T_i} = \frac{\lambda h}{4A_e}\left[(b_i^2 + c_i^2)T_i + (b_ib_j + c_ic_j)T_j + (b_ib_m + c_ic_m)T_m\right]$$

For derivatives with respect to T_j and T_m, replace row index i with j and m.

The derivative is a linear combination of nodal temperatures.
Its coefficients form a row of the conduction matrix.

Minimization: element conduction matrix

1. Coefficient of each nodal temperature

$$K_{\lambda,ab}^{(e)} = \frac{\lambda h}{4A_e}(b_a b_b + c_a c_b), \qquad a, b \in \{i, j, m\}$$

Index a denotes the row; index b denotes the column.

2. Three derivatives give a 3 × 3 matrix

$$\mathbf{K}_{\lambda,e} = \frac{\lambda h}{4A_e} \begin{pmatrix} b_i^2 + c_i^2 & b_i b_j + c_i c_j & b_i b_m + c_i c_m \\ b_j b_i + c_j c_i & b_j^2 + c_j^2 & b_j b_m + c_j c_m \\ b_m b_i + c_m c_i & b_m b_j + c_m c_j & b_m^2 + c_m^2 \end{pmatrix}$$

3. Properties of the conduction matrix

The matrix is symmetric and depends on geometry, thickness, and thermal conductivity.

Conduction contribution to the functional derivative: $\boldsymbol{\partial \Phi_{\lambda,e} / \partial \mathbf{T}_e = \mathbf{K}_{\lambda,e} \mathbf{T}_e}$

Minimization: source contribution to the right-hand side

1. Scalar expression

$$\Phi_{Q,e} = -\frac{Q_e h A_e}{3}(T_i + T_j + T_m)$$

2. Derivative with respect to each nodal temperature

$$\frac{\partial \Phi_{Q,e}}{\partial T_i} = -\frac{Q_e h A_e}{3}, \qquad \frac{\partial \Phi_{Q,e}}{\partial T_j} = -\frac{Q_e h A_e}{3}, \qquad \frac{\partial \Phi_{Q,e}}{\partial T_m} = -\frac{Q_e h A_e}{3}$$

3. Source load vector

$$\frac{\partial \Phi_{Q,e}}{\partial \mathbf{T}_e} = -\mathbf{F}_{Q,e}, \qquad \boxed{\mathbf{F}_{Q,e} = \frac{Q_e h A_e}{3}\begin{pmatrix}1\\1\\1\end{pmatrix}}$$

The source affects only the right-hand side.
Sum of nodal loads: $FQ_{,i} + FQ_{,j} + FQ_{,m} = Q_e h A_e$.

Minimization: prescribed outward heat flux

1. Scalar expression for edge i, j

$$\Phi_{q,\gamma} = \frac{\bar{q}_n h L}{2}(T_i + T_j)$$

2. Derivatives at the two edge nodes

$$\frac{\partial \Phi_{q,\gamma}}{\partial T_i} = \frac{\bar{q}_n h L}{2}, \qquad \frac{\partial \Phi_{q,\gamma}}{\partial T_j} = \frac{\bar{q}_n h L}{2}$$

3. The load and derivative have opposite signs

$$\frac{\partial \Phi_{q,\gamma}}{\partial \mathbf{T}_\gamma} = -\mathbf{F}_{q,\gamma}, \qquad \boxed{\mathbf{F}_{q,\gamma} = -\frac{\bar{q}_n h L}{2}\begin{pmatrix}1\\1\end{pmatrix}}$$

Inward heat flux $q_{in} > 0$ gives a positive load.

Type II conditions affect only the right-hand side.
Outward flux gives a negative load; inward flux gives a positive load.

Minimization: derivatives of the convection contribution

1. Complete scalar expression

$$\Phi_{\alpha,\gamma} = \frac{\alpha h L}{6}\left[T_i^2 + T_i T_j + T_j^2 - 3T_\infty(T_i + T_j) + 3T_\infty^2\right]$$

2. Derivative with respect to T_i

$$\frac{\partial \Phi_{\alpha,\gamma}}{\partial T_i} = \frac{\alpha h L}{6}(2T_i + T_j - 3T_\infty) = \frac{\alpha h L}{6}(2T_i + T_j) - \frac{\alpha h L T_\infty}{2}$$

3. Derivative with respect to T_j

$$\frac{\partial \Phi_{\alpha,\gamma}}{\partial T_j} = \frac{\alpha h L}{6}(T_i + 2T_j - 3T_\infty) = \frac{\alpha h L}{6}(T_i + 2T_j) - \frac{\alpha h L T_\infty}{2}$$

The derivative of the constant term $\alpha h L T_\infty^2/2$ is zero.

Each derivative contains a part involving unknown temperatures
and a known part depending on the ambient temperature.

Minimization: convection matrix and ambient load

1. The two derivatives together

$$\begin{pmatrix} \partial \Phi_{\alpha,\gamma}/\partial T_i \\ \partial \Phi_{\alpha,\gamma}/\partial T_j \end{pmatrix} = \frac{\alpha h L}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \end{pmatrix} - \frac{\alpha h L T_\infty}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2. Matrix and load vector

$$\boxed{\mathbf{K}_{\alpha,\gamma} = \frac{\alpha h L}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}}, \qquad \boxed{\mathbf{F}_{\alpha,\gamma} = \frac{\alpha h L T_\infty}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

3. Compact form

$$\frac{\partial \Phi_{\alpha,\gamma}}{\partial \mathbf{T}_\gamma} = \mathbf{K}_{\alpha,\gamma} \mathbf{T}_\gamma - \mathbf{F}_{\alpha,\gamma}$$

Type III conditions affect the matrix and right-hand side. At $T_i = T_j = T\infty$, $K\alpha,\gamma T\gamma = F\alpha,\gamma$, so the convection contribution to the derivative is zero.

Combined contributions of one finite element

1. Edge i, j embedded in node order (i, j, m)

$$\mathbf{K}_{\alpha,e}^{(ij)} = \frac{\alpha h L}{6}\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \mathbf{F}_{q,e}^{(ij)} = -\frac{\bar{q}_n h L}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{F}_{\alpha,e}^{(ij)} = \frac{\alpha h L T_\infty}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Convection (Type III) | Prescribed heat flux (Type II) | Ambient temperature

2. Sum of all element contributions

$$\mathbf{K}_e = \mathbf{K}_{\lambda,e} + \sum_{\gamma \in E_{3,e}} \mathbf{K}_{\alpha,e}^{(\gamma)}, \qquad \mathbf{F}_e = \mathbf{F}_{Q,e} + \sum_{\gamma \in E_{2,e}} \mathbf{F}_{q,e}^{(\gamma)} + \sum_{\gamma \in E_{3,e}} \mathbf{F}_{\alpha,e}^{(\gamma)}$$

3. Vector of local derivatives

$$\mathbf{R}_e = \frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \mathbf{K}_e \mathbf{T}_e - \mathbf{F}_e$$

Only one boundary-condition type applies on an edge.
The local residual R_e need not vanish on its own.

General case: five terms in the local derivative

Internal source, Type III on edge i, j, Type II on edge j, m

$$\mathbf{R}_e = \frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \frac{\lambda h}{4A_e} \begin{pmatrix} b_i^2 + c_i^2 & b_i b_j + c_i c_j & b_i b_m + c_i c_m \\ b_j b_i + c_j c_i & b_j^2 + c_j^2 & b_j b_m + c_j c_m \\ b_m b_i + c_m c_i & b_m b_j + c_m c_j & b_m^2 + c_m^2 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

$$- \frac{Q_e h A_e}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\bar{q}_n h L_{jm}}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \frac{\alpha h L_{ij}}{6} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix} - \frac{\alpha h L_{ij} T_\infty}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Compact form

$$\mathbf{R}_e = (\mathbf{K}_{\lambda,e} + \mathbf{K}_{\alpha,e})\mathbf{T}_e - \mathbf{F}_{Q,e} - \mathbf{F}_{q,e} - \mathbf{F}_{\alpha,e}$$

Only the sum of element contributions vanishes
after global assembly at free nodes.

Global assembly: nodal temperature derivatives

1. Global functional

$$\Phi(\mathbf{T}) = \sum_{e=1}^{n_e} \Phi_e, \qquad \mathbf{T} = \underbrace{(T_1 \cdots T_i \cdots T_j \cdots T_m \cdots T_N)^{\mathsf{T}}}_{N\times 1}$$

N: number of mesh nodes; n_e: number of elements. A shared node has one temperature.

2. Assembly of derivatives at all nodes

$$\underbrace{\frac{\partial \Phi}{\partial \mathbf{T}}}_{N\times 1} = \begin{pmatrix} \partial\Phi/\partial T_1 \\ \vdots \\ \partial\Phi/\partial T_N \end{pmatrix} = \sum_{e=1}^{n_e} \begin{pmatrix} \partial\Phi_e/\partial T_1 \\ \vdots \\ \partial\Phi_e/\partial T_N \end{pmatrix} = \mathbf{R}$$

3. Contributions occur only at element nodes

$$\frac{\partial \Phi_e}{\partial T_n} = \begin{cases} R_{e,n}, & n \in \{i,j,m\}, \\ 0, & n \notin \{i,j,m\}. \end{cases}$$

Contributions at i, j, m may be nonzero.
At a shared node, adjacent element contributions add.

Minimum condition: $R_a = 0$ only at free nodes. Prescribed temperatures remain fixed.

Global assembly: matrix and load vector

1. Element contributions mapped to global node numbers

$$\underbrace{\frac{\partial \Phi}{\partial \mathbf{T}}}_{N\times 1} = \Bigg[\underbrace{\begin{pmatrix} K_{11}^{(1)} & K_{12}^{(1)} & K_{13}^{(1)} & 0 & \cdots & 0 \\ K_{21}^{(1)} & K_{22}^{(1)} & K_{23}^{(1)} & 0 & \cdots & 0 \\ K_{31}^{(1)} & K_{32}^{(1)} & K_{33}^{(1)} & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}_{N\times N}}_{\tilde{\mathbf{K}}^{(1)}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & K_{22}^{(2)} & K_{23}^{(2)} & K_{24}^{(2)} & \cdots & 0 \\ 0 & K_{32}^{(2)} & K_{33}^{(2)} & K_{34}^{(2)} & \cdots & 0 \\ 0 & K_{42}^{(2)} & K_{43}^{(2)} & K_{44}^{(2)} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}_{N\times N}}_{\tilde{\mathbf{K}}^{(2)}} + \cdots \Bigg] \begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ \vdots \\ T_N \end{pmatrix}$$

$$- \Bigg[\underbrace{\begin{pmatrix} F_1^{(1)} \\ F_2^{(1)} \\ F_3^{(1)} \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{N\times 1}}_{\tilde{\mathbf{F}}^{(1)}} + \underbrace{\begin{pmatrix} 0 \\ F_2^{(2)} \\ F_3^{(2)} \\ F_4^{(2)} \\ \vdots \\ 0 \end{pmatrix}_{N\times 1}}_{\tilde{\mathbf{F}}^{(2)}} + \cdots \Bigg] = \mathbf{R}.$$

At shared nodes, add contributions at matching global positions.

$$\frac{\partial \Phi}{\partial \mathbf{T}} = \underbrace{\left(\tilde{\mathbf{K}}^{(1)} + \tilde{\mathbf{K}}^{(2)} + \cdots + \tilde{\mathbf{K}}^{(n_e)}\right)}_{\mathbf{K}} \mathbf{T} - \underbrace{\left(\tilde{\mathbf{F}}^{(1)} + \tilde{\mathbf{F}}^{(2)} + \cdots + \tilde{\mathbf{F}}^{(n_e)}\right)}_{\mathbf{F}} = \mathbf{K}\mathbf{T} - \mathbf{F} = \mathbf{R}.$$

At unknown-temperature nodes, $R_a = 0$. Apply prescribed temperatures separately.

Dirichlet boundary conditions in the global system

1. Free and prescribed temperatures

$$\mathbf{T} = \begin{pmatrix} \mathbf{T}_f \\ \overline{\mathbf{T}}_c \end{pmatrix}, \qquad N = N_f + N_c$$

f: free nodes; c: nodes with prescribed temperatures.

2. Free-node rows of the assembled system

$$\frac{\partial \Phi}{\partial \mathbf{T}_f} = \mathbf{K}_{ff}\mathbf{T}_f + \mathbf{K}_{fc}\overline{\mathbf{T}}_c - \mathbf{F}_f = \mathbf{0}$$

3. Known temperatures moved to the right-hand side

$$\underbrace{\mathbf{K}_{ff}}_{N_f \times N_f}\ \underbrace{\mathbf{T}_f}_{N_f \times 1} = \underbrace{\mathbf{F}_f - \mathbf{K}_{fc}\overline{\mathbf{T}}_c}_{N_f \times 1}$$

Dirichlet conditions modify the unknown-node system, without adding an integral to the functional.

Special case: no convection

1. Set α = 0 on the former Robin boundaries

$$\alpha = 0 \quad \Longrightarrow \quad \Phi^{(\alpha)} = 0, \quad \mathbf{K}^{(\alpha)} = \mathbf{0}, \quad \mathbf{F}^{(\alpha)} = \mathbf{0}$$

Here $q_n = 0$: thermal insulation, not a new prescribed temperature.

---

2. Conduction, heat source, and prescribed flux remain

$$\Phi_e = \frac{\lambda h}{8A_e}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right]$$
$$- \frac{QhA_e}{3}(T_i + T_j + T_m) + \sum_{\gamma=(p,s)\in\mathcal{E}_{2,e}} \frac{\bar{q}_{n,\gamma}hL_\gamma}{2}(T_p + T_s).$$

3. Dirichlet and Neumann conditions remain

$$\mathbf{K} = \mathbf{K}^{(\lambda)}, \qquad \mathbf{F} = \mathbf{F}^{(Q)} + \mathbf{F}^{(q)}$$

Prescribed temperatures on the original Dirichlet boundaries remain unchanged.

Special case: zero prescribed heat flux

1. After setting $\alpha = 0$, set $\bar{q}_n = 0$

$$\bar{q}_n = 0 \quad \Longrightarrow \quad \Phi^{(q)} = 0, \qquad \mathbf{F}^{(q)} = \mathbf{0}$$

The former Neumann boundaries also become thermally insulated.

---

2. Prescribed temperatures and the volumetric source remain

$$\Phi_e = \frac{\lambda h}{8A_e}\left[(b_i T_i + b_j T_j + b_m T_m)^2 + (c_i T_i + c_j T_j + c_m T_m)^2\right] - \frac{QhA_e}{3}(T_i + T_j + T_m)$$

$$\mathbf{K}_{ff}^{(\lambda)}\mathbf{T}_f = \mathbf{F}_f^{(Q)} - \mathbf{K}_{fc}^{(\lambda)}\overline{\mathbf{T}}_c$$

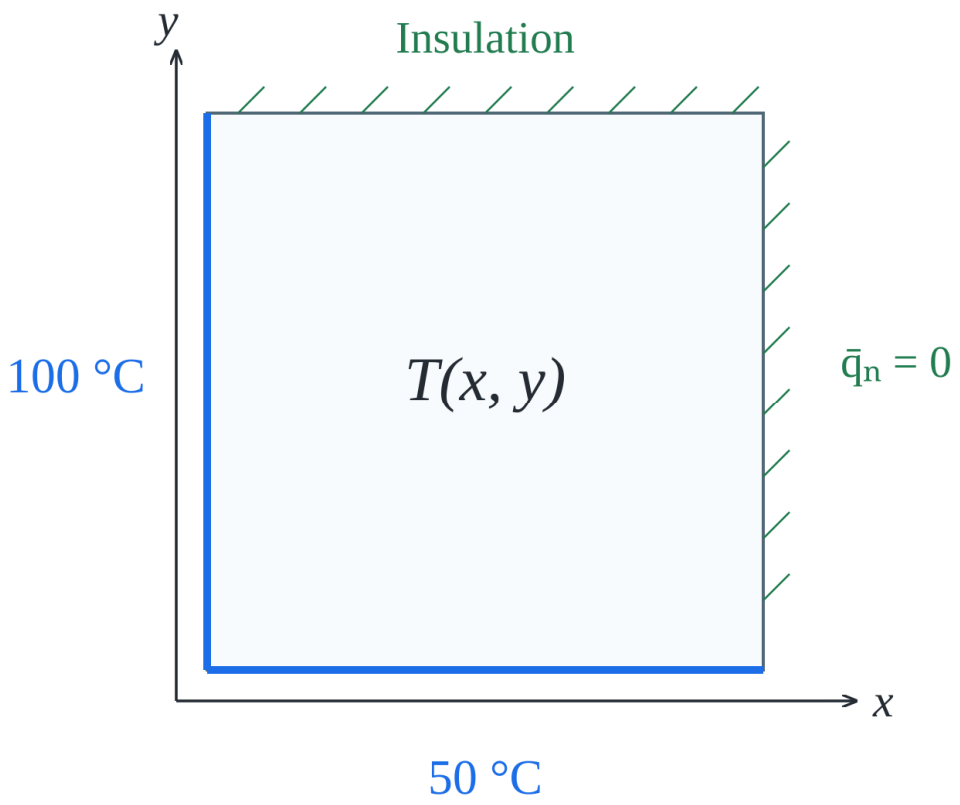
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3. If there is no internal heat source, also set $Q = 0$

$$\mathbf{K}_{ff}^{(\lambda)}\mathbf{T}_f = -\mathbf{K}_{fc}^{(\lambda)}\overline{\mathbf{T}}_c$$

Even for Q = 0, the free-node right-hand side includes the prescribed-temperature contribution.

Teaching example: plate with prescribed temperatures



1. Dimensions and material properties

$$L_x = L_y = 5\,\mathrm{mm}, \quad h = 1\,\mathrm{mm}$$

$$\lambda = 45\,\mathrm{W/(m \cdot K)}, \quad Q = 0$$

2. Boundary conditions

Left: T = 100 °C.
Bottom: T = 50 °C.
Right and top: $q_n = 0$.

3. Node at the boundary intersection

Assign one value to the corner node.
The original calculation uses 50 °C.

This original example uses Dirichlet conditions. Its boundary arrangement is specified separately.

Solution and temperature field recovery

1. Solution for free nodal temperatures

$$\mathbf{K}_{ff}\mathbf{T}_f = \mathbf{F}_f - \mathbf{K}_{fc}\overline{\mathbf{T}}_c$$

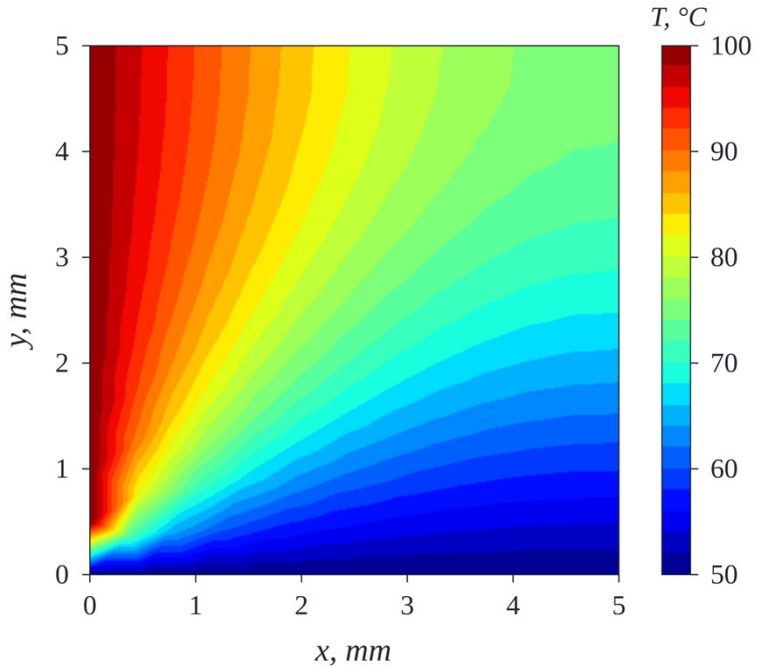
Add the known values Tc to obtain temperatures at all mesh nodes.

2. Temperature recovery within an element

$$T^{(e)}(x,y) = N_i(x,y)T_i + N_j(x,y)T_j + N_m(x,y)T_m$$

Three nodal temperatures and shape functions define the field within the triangle.

3. Temperature field of the whole plate



Nodal temperatures → element fields → temperature field of the whole plate.



Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

CONCLUSION